Abdelrahman Elewah

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Summary

AI Integration Lead and Machine Learning Engineer specializing in designing and deploying scalable AI systems across enterprise and research environments. Currently leading AI and automation integration initiatives at MPAC, modernizing property assessment workflows through LLMs, RAG pipelines, and large vision models. Expertise in blueprint understanding, document intelligence, semantic search, and real-time IoT data frameworks. Proven ability to translate research-grade AI into production-ready systems, aligning technical innovation with organizational KPIs. Strong foundation in software engineering, cloud-native architectures, and cross-functional leadership driving measurable digital transformation.

Technologies

LLM Frameworks & Tools: LangChain - LangGraph - LangSmith - RAG - Agentic AI - Docling - MinerU - MCP

Programming Languages: Python - C/C++ - JavaScript

Web Technologies: REST APIs - React - HTML - CSS - Bootstrap - FastAPI - Flask

Automation Tools: Jenkins - GitLab CI/CD - GitHub Actions Workflow

Cloud & Deployment: AWS EC2 - AWS App Runner - Elastic Container Service - Elastic Beanstalk - CI/CD Pipelines

Database Systems: PostgreSQL - MongoDB - MySQL - NoSQL - SQL - Spark - Hadoop

DevOps Tools: Docker - Docker Compose - Kubernetes - Dev Containers

Development Environments: GitHub - GitLab - VS Code - Anaconda

Experience

Artificial Intelligence Integration Lead, MPAC — Pickering, ON — Jul 2025 – Present

- Lead enterprise-wide integration of **AI and automation solutions** across property assessment workflows, modernizing MPAC's digital infrastructure and operational efficiency.
- Architect and implement scalable **data pipelines** using Python, LangGraph, FastAPI, Docling, MinerU, PostgreSQL, and MCP servers to support intelligent **building plan parsing and document understanding**.
- Design and deploy advanced **RAG (Retrieval-Augmented Generation)** systems for building plan understanding and semantic search aligned with MPAC assessment standards.
- Integrate and optimize **large vision-language models (e.g., Qwen3-VL, LayoutLMv3, PaddleOCR-VL, PP-StructureV3)** and other detection and segmentation models such as **RF-DETR** for architectural object detection, spatial reasoning, document layout understanding, and semantic segmentation.
- Lead technical documentation, stakeholder presentations, and cross-team **knowledge transfer initiatives** to ensure sustainable AI adoption.

Instructional Specialist (part-time), 2U / University of Toronto — Remote — Jan 2023 – Apr 2025

- Contributed to the success of the **University of Toronto** 's online **Data Analytics Boot Camp**, supporting 100+ learners in mastering practical skills for data-driven careers.
- Facilitated hands-on workshops in **Python, Database, Machine Learning, and Data Visualization**, resulting in a **15% increase** in student satisfaction scores.
- Supported the deployment of **real-world capstone projects**, helping learners apply techniques in domains such as **healthcare, HR, and finance**.

Research Assistant, Ontario Tech University — Oshawa, ON — Jan 2020 – Apr 2025

- Designed the **SensorsConnect** framework, enabling real-time **IoT device interoperability** and scalable data exchange, inspired by the principles of the **World Wide Web**. Publication
- Developed an **Agentic IoT Search Engine (ASE-IoT)** that integrates **LLMs**, **RAG**, and autonomous agents to enable **natural language querying** of live IoT data, enhancing search precision and user interaction. publication

- Co-led an **OVIN-funded project** to develop **autonomous vehicle curriculums** and conduct applied research on **software-defined vehicles (SDVs)** and **digital twins**, contributing to workforce development in emerging mobility technologies.
- Collaborated with **Eagle Aerospace** to prototype an **Aircraft Deceleration Early Warning System**, enhancing runway safety through predictive analytics and early alert mechanisms.

Back-End Developer(part-time) Tamra-IoT, Toronto, ON
May 2019 – Jan 2023

- **Architected** secure and scalable **IoT platforms** by integrating **MQTT over TLS**, **cloud infrastructure**, and **mobile control interfaces**, enhancing real-time communication and remote device management.
- **Collaborated** on **business management** and strategic planning, contributing to key decisions that optimized resource allocation, improved product direction, supported **mentorship**, and accelerated go-to-market execution.
- **Developed a curriculum** to teach **IoT concepts** to high school students, promoting early STEM engagement and empowering the next generation with practical, hands-on IoT experience.
- **Designed** and deployed **Over-The-Air (OTA) firmware update mechanisms** and implemented **robust IoT device management systems**, significantly reducing maintenance costs and improving system resilience. Publication

Education

PhD **Ontario Tech University**, Electrical and Computer Engineering — Jan 2020 – Mar 2025

- GPA: 4.22/4.3 Link to Transcript issued by Ontario Tech University
- **Coursework:** Real-Time Data For IoT, Communication Networks, Knowledge Discovery & Data Mining, Data Visualizations
- **Thesis**: SensorsConnect: World Wide Web for Internet of Things.

MSc **Benha University**, Electrical Engineering — Feb 2013 – Jan 2018
BSc **Benha University**, Electrical Engineering — Sep 2008 – Jun 2012

Selected Projects

Localelive: Agentic Search Engine for Real-Time IoT Data Live Demo publication — github.com/repo

- **Developed** a real-time IoT search engine powered by **LLMs and RAG**, enabling users to query complex sensor data using natural language, improving query efficiency and decision making.
- **Implemented** a semantic search pipeline using **Sentence-BERT and HNSW indexing**, reducing query latency by 73% and enhancing relevance in top-k retrieval across diverse IoT datasets.
- **Managed** over 37,000 real-time IoT documents from 500+ service types in **MongoDB** with geo-indexing, ensuring scalable and location-aware data access for time-sensitive decision-making.
- **Achieved** 92% top-1 accuracy in complex intent detection and information retrieval, surpassing systems like Gemini, and significantly improving user satisfaction and task completion rates.
- **Applied** in real-time urban scenarios—such as locating least-crowded clinics, nearest available parking, and lowest gas prices—demonstrating direct utility for smart city applications.
- **Technologies:** FastAPI, Leveraged LangGraph, Tavily API, OpenRouteService, VectorDB, and Sentence-BERT to build a modular and extensible architecture for dynamic IoT data exploration and retrieval.
- **Deployed** the system in **AWS** using **Docker Compose and Traefik** Traefik live dashboard, enabling seamless container orchestration, automated HTTPS provisioning, and scalable reverse proxy management for reliable production-grade deployment.

Story-to-Movie Recommender Chatbot (RAG-based) Live demo — github.com/repo

- Developed a **retrieval-augmented generation (RAG)** system combining vector search with LLMs to deliver context-aware answers, reducing hallucination rates by **30%**.
- Built a **semantic search pipeline**, enabling retrievals from 1K+ documents.
- Fine-tuned user prompts and applied **prompt chaining** techniques to improve answer relevance, validated through user feedback and precision metrics.
- Leveraged **Pandas**, **OpenAI API**, and **Tenacity** to ensure resilient API usage and robust data handling under real-time loads.

Apply Lightweight Fine-Tuning to a Foundation Model github.com/repo

- **Built** an end-to-end NLP pipeline using **PyTorch**, **TensorFlow** and **Hugging Face Transformers**: loaded a pre-trained **GPT-2** model and prepared the **AG News** dataset for news classification.
- **Applied** parameter-efficient fine-tuning (PEFT) using **LoRA adapters** to fine-tune GPT-2 while keeping the base model's weights frozen, **reducing training time and memory usage by over 60% compared to full fine-tuning**.
- **Achieved** a significant improvement: **boosted accuracy from 83.16% to 88.95%** on the AG News dataset using LoRA-fine-tuning, **demonstrating the effectiveness of PEFT in enhancing model performance with minimal compute**.

RadViz-Plotly github.com/repo

- **Developed RadViz-Plotly**, an **open-source Python package** that generates **2D and 3D Radial Visualization (RadViz)** plots for **high-dimensional datasets**, enabling broader accessibility to dimensionality reduction techniques in research and industry. publication
- **Enabled data scientists** and analysts to explore and interpret **complex data distributions** interactively using **Plotly**, significantly improving **model explainability** and **decision-making** in analytics workflows.
- **Facilitated** deeper insights into **high-dimensional data** by revealing **hidden clusters, outliers, and trends**, increasing user engagement through intuitive visual interfaces.